

ABDN Fellowship Programme

Electrophysiology Module

Curriculum Proposal

July – December 2026 | Virtual (Zoom) | 5 Months

Teaching Assistants

Nicole Vissers (Lead)
Happiness Eric Aigbogun

Programme Details

Duration: July – December 2026
Format: Weekly virtual sessions (Zoom)
Expected participants: ~10

Programme Overview

This curriculum covers the foundations and practical applications of electrophysiology for early-career African neuroscientists. It is designed to guide participants from introductory neuroscience concepts to recording techniques, experimental design, data analysis, and paper writing, while keeping available resources into consideration.

A central goal of the course is to empower African scientists with practical knowledge and accessible tools that they can use to advance their own scientific careers independently during and after the fellowship. Participants will receive an introduction to neuroscience and electrophysiological recording methods, learn how neural data can be modeled and analyzed, and explore open-source tools and resources that make neuroscience research more accessible and reproducible.

The curriculum also emphasizes the broader skills required to become an independent researcher, including how to identify and refine research questions, design experiments, navigate ethical considerations, responsibly use AI tools in research, and find appropriate datasets for analysis. In addition, participants will learn strategies for finding and evaluating scientific literature, building collaborative networks, seeking mentorship or technical support when challenges arise, and engaging with the wider neuroscience community.

Finally, the course introduces practical aspects of scientific publishing, including selecting an appropriate journal, understanding the publication process, and developing effective scientific writing skills to communicate research findings clearly and professionally.

Focus

Developing practical, transferable skills that participants can continue to apply independently after the fellowship. The programme focuses on empowering early-career researchers to take initiative in designing research questions, working with open-source neuroscience tools and datasets, analysing electrophysiological data, building collaborative networks, and communicating their findings through scientific writing and publication.

Programme Goals

By the end of the programme, participants will be able to:

- Understand core principles of systems neuroscience and electrophysiology
- Design basic neuroscience experiments with appropriate controls and structure
- Acquire, preprocess, and analyse electrophysiological data using open-source tools
- Identify and critically evaluate scientific literature
- Formulate a research question and align it with available datasets
- Produce structured scientific writing suitable for publication
- Navigate the peer review and journal submission process
- Use AI tools responsibly for coding, writing, and research support
- Independently continue research using open datasets and open tools

Tools & Platforms Used Across the Programme

- OpenNeuro.org – African electrophysiology open datasets
- EBRAINS – European brain data infrastructure with African-relevant datasets\
- International Brain Laboratory
- DANDI

- Google Colab – accessible cloud Python environment (no local install required)
- DeepUnitMatch
- Bombcell
- Pynapple (FlatIron Institute)

- NEURON
- Brian
- NetPyNE
- NEST
- Human Neocortical Neurosolver
- MNE-Python, SciPy, NumPy, Matplotlib, Pandas
- Zoom + shared Google Drive folder for all session materials

Assessment Structure

During the fellowships students will be asked to perform several assessments to build towards a final publication.

Assessment	When
Weekly reflection (short written)	End of each week
Research question & design	End of Module 2
Mini analysis assignment	End of Module 3
Literature review draft	End of Module 4
Final paper / submission-ready draft	End of Module 5

Module 1: Introduction to Electrophysiology (July)

Learning Objective: Understand the electrical properties of neurons and the principles underlying electrophysiological measurement.

Week	Topic	Key Concepts	Tools / Resources	Assessment
Ask the students to prepare a few slides to introduce themselves (roughly 5 minutes)				
Week 1 18th of July Nicole	Introduction to each other - Meet the Mentees	Show our data & brainstorm on what participants want to do; discuss social rules and expectations; explore background knowledge of participants; set up WhatsApp community for ongoing communication	Shared Google Drive; WhatsApp Community; Zoom	Participation in class and student self-introductions (5-minute slide presentation per student)
Week 2 23rd of July Nicole	Brief introduction to Neuroscience	Anatomy of a neuron; synaptic transmission; membrane potentials; action potential generation and propagation; types of neurons; overview of how the brain processes information	Neuroscience Online (free); 3D Brain App; OpenStax Neuroscience	Reflection: What surprised you most about neuron structure?
Week 3 30th of July Nicole	Recording techniques used in Neuroscience	Patch clamp (whole-cell, cell-attached); extracellular recordings; EEG and MEG; local field potentials (LFPs); multi-unit activity (MUA); EEG electrode placement (10-20 system); signal-to-noise ratio; overview of Neuropixels probes	Virtual patch clamp simulator (U. Michigan); MATLAB/Python demo; pClamp conceptual walkthrough; MNE-Python (first hands-on); OpenNeuro African EEG dataset; Neuropixels 2025 course: https://www.ucl.ac.uk/brain-sciences/research/neuropixels/training/2025-neuropixels-course (Student to watch)	Pick your preferred scientific recording technique and justify why, 1 paragraph

Give them this to watch:

<https://www.ucl.ac.uk/brain-sciences/research/neuropixels/training/2025-neuropixels-course>

Week 4 6th of August Invite Speaker? Developers of software? Nicole	Introduction to computational models of neural systems	Why computational modelling? Overview of neural modelling approaches; integrate-and-fire models; Hodgkin-Huxley model; network models; free online resources for modelling; how models complement experimental data	NEURON, Brian, NetPyNE, Nest Simulator, Human Neocortical Neurosolver	Short modelling exercise using a beginner-friendly tutorial in NEURON or Brian
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Module 2: Experimental Design (August)

Learning Objective: How to navigate literature,

Week	Topic	Key Concepts	Tools / Resources	Assessment
Week 1 15th of August Happines s	Introduction to searching Literature	PubMed/Google Scholar search strategies; Boolean operators; evaluating source quality and determining whether a paper can be trusted; citation managers; using Connected Papers to map a field visually	PubMed; Google Scholar; managing them through Zotero, Connected Papers	Find a few topics you are interested in and bring 3–5 papers to discuss next session
Week 2 22th of August Happines s	Experimental Design & finding a research question; group students into 3 teams based on interests	Controls and blinding; sample size for electrophysiology; ethical considerations in study design; data management planning; what animal models exist and why to choose one over another; formulating a focused research question; group students into 3 teams based on interests	BIDS Validator; OpenNeuro upload walkthrough; shared experimental design template	Draft a one-page experimental design for a proposed study, including research question, model/dataset choice, and ethical considerations
Week 3 29th of August Invite speaker & European Animal	Ethics & AI Usage	What ethical rules apply when designing experiments (human and animal); informed consent; IRB processes in African institutions; the 3Rs framework; how to use LLMs	LLMs: ChatGPT, Claude (free tiers); WHO ethics guidelines; example African IRB	Revisit your experimental design from Week 2 – consider your LLM usage strategy and identify any

Research Association?		responsibly in research (coding support, writing assistance, literature summaries); what assumptions do LLMs make about data; understanding LLM limitations and hallucinations; practical demos with ChatGPT and Claude	consent templates	ethical considerations specific to your proposed study
Happiness				
Week 4 5th of September Invite speakers from open source? Happiness	Find right dataset	How to find the right dataset to answer your research question; overview of major open neuroscience data repositories; understanding data formats (NWB, BIDS, EDF); evaluating dataset quality and relevance; downloading and exploring a dataset; how to contribute to and navigate community datasets	IBL (internationalbrainlab.com); DANDI (dandiarchive.org); eBRAIN; OpenNeuro; Pynapple for data loading; Allen Neural Dynamics OpenScope Community: https://allenneuraldynamics.github.io/openscope-community-predictive-processing/how_to_contribute/	Find a dataset that would be useful to answer the research question we have developed, write a brief justification (half a page) including why this dataset fits your question
Week 5 10th of September Nicole	How to reach out to people if you get stuck	What to do if you get stuck on a technical problem or dataset; who to reach out to and how; writing professional scientific emails; finding help through forums, GitHub issues, Slack communities, and LinkedIn; building a support network as an early-career African scientist	LinkedIn; ResearchGate; GitHub Issues; Neurostars forum; email templates	Write two example emails: one to a dataset owner asking a technical question, and one to a potential mentor or collaborator

Module 3: Scientific Writing – Literature Review (September)

Learning Objective: Develop skills in scientific literature synthesis and begin drafting a structured literature review focused on a neural signalling topic relevant to African populations.

Where possible, participants are encouraged to target African neuroscience journals (e.g. African Journal of Neurological Sciences) or post to AfricArXiv as a preprint to maximise visibility and impact on the continent.

Week	Topic	Key Concepts	Tools / Resources	Assessment
Week 1 17th of September Invite Editor and Reviewer (Elizabeth Bik? Shari Wiseman?) Nicole	Process of picking a journal and publishing a paper	How the publishing process works; how to pick the right journal for your topic; open access vs subscription journals; understanding impact factor and scope; author guidelines and formatting requirements; what the submission and peer review process looks like; how to find and read reviewing comments	JANE (Journal/Auth or Name Estimator); journal websites; open access reviewing comments; AfricArXiv	Pick a journal that best fits your current research project; write down 3 potential questions reviewers might ask you
Week 2 24th of September Speaker? Nicole	Reading & Synthesising Papers	How to read a scientific paper efficiently (abstract → methods → results → discussion); identifying gaps and contradictions in the literature; grouping papers by theme; building a synthesis matrix; using Connected Papers to visualise literature networks	Connected Papers (connectedpapers.com); Zotero; synthesis matrix template (shared Google Sheet)	Complete a synthesis matrix for your 10 references, identifying at least 2 research gaps
Week 3 1st of October Nicole	Structuring the Literature Review	Introduction, thematic body sections, transitions, discussing conflicting evidence, conclusion and research gap; how to structure an argument; writing for a reader who is not a specialist in your exact topic	Google Docs; shared review template; example African neuroscience papers	Submit a full outline + written introduction paragraph (250–300 words)
Week 4 10th of October Nicole not there! Happiness	Writing the First Draft	Scientific tone and voice; definitive vs hedged language, say what the data shows, not what it appears to show; paraphrasing vs quoting; in-text citation style (APA or Vancouver); presenting figures and results clearly; writing a strong discussion	Google Docs; Grammarly; Zotero citation plugin	Submit full first draft of literature review

SUBMIT REVIEW PAPER!

Module 4: Data Analysis (October)

Learning Objective: Analyze electrophysiological signals using open-source tools, with a focus on reproducible, FAIR-compliant pipelines applied to African brain data.

Week	Topic	Key Concepts	Tools / Resources	Assessment
Week 1 17th of October Happiness	Spike Detection	Action potential detection algorithms; threshold vs template matching; spike sorting concepts (KiloSort overview); DeepUnitMatch and Bombcell for quality control; single-unit vs multi-unit activity; interpreting spike trains	MNE-Python; SciPy; OpenNeuro African dataset	Run spike detection on your chosen dataset; report false positive rate and describe the quality of the output
Week 2 24th of October Happiness	Signal Filtering	Low-pass, high-pass, bandpass, notch filters and when to use each; EEG frequency bands (delta, theta, alpha, beta, gamma); artefact identification (eye blinks, muscle noise, power line interference); ICA for artefact removal; power spectral density (PSD) plots	MNE-Python (filter + ICA); Matplotlib for PSD plots	Apply a bandpass filter and ICA to your dataset; compare PSD plots before and after processing
Week 3 31st of October Happiness	Data Interpretation, ERPs & dimensionality reduction	Epoching: cutting continuous data into event-locked segments; baseline correction; event-related potentials (P300, N200) and their significance; averaging across trials; statistical thresholding; Pynapple for time series analysis; dimensionality reduction methods: Principal Component Analysis (PCA) for identifying dominant patterns in population	MNE-Python (epochs + ERP); Pynapple; Scikit-learn (PCA, LDA); dPCA Python package; statsmodels (GLMs); African/IBL dataset event files; Matplotlib	Full pipeline mini-assignment: raw signal → filter → ICA → epoch → ERP plot → apply PCA to your epoched data → written

		activity; Linear Discriminant Analysis (LDA) for separating neural states; demixed PCA (dPCA) for separating task-related variance in neural population data; Generalised Linear Models (GLMs) for modelling spike rates as a function of experimental variables; interpreting results in the context of your research question		interpretation (1 page)
Week 4 7th of November Happiness	Software Tools Deep Dive & Reproducibility workflows	Building a fully documented reproducible analysis pipeline; Jupyter notebooks for transparent step-by-step analysis; version control with GitHub: commits, branches, README files; BIDS output formatting and validation; data sharing on OpenNeuro or DANDI; writing a methods section that fully enables replication	Jupyter; GitHub; BIDS Validator; OpenNeuro; DANDI; Google Colab, OpenScope	Submit a fully documented Jupyter notebook covering your complete September analysis pipeline, with inline comments and a short replicable methods paragraph

Module 5: Finalizing the Data Paper (November)

Learning Objective: Revise, peer-review, and produce a submission-ready draft of the literature review or short research paper, ready for journal submission or posting as a preprint.

Week	Topic	Key Concepts	Tools / Resources	Assessment
Week 1 12th of November Nicole	Peer Review Within the Group	How to give constructive feedback; structured peer review using a rubric – assessing clarity, methods rigour, results presentation, and discussion quality; understanding what journal peer reviewers look for; responding to reviewer comments professionally and point-by-point	Shared peer review rubric; Google Docs comments	Submit a structured peer review of a fellow participant's draft using the shared rubric
Week 2 19th of November Nicole	Revision & Responding to Feedback	Tracking changes; prioritising revisions: major vs minor comments; rewriting for clarity and concision; responding to comments point-by-point; ensuring methods are reproducible and fully described; updating figures and captions	Google Docs track changes; Grammarly	Submit revised draft with response-to-reviewer notes
Week 3 26th of November Nicole	Continue to work on your paper	Further revision and polishing; addressing remaining reviewer comments; improving figures; strengthening the discussion; ensuring citation consistency and formatting; abstract writing and title optimisation	Google Docs; Grammarly; Zotero; target journal author guidelines	Submit journal-formatted draft + chosen target journal rationale

Week 4
5th of
December

Together!

**Final Submission &
Programme
Wrap-Up**

Final polish; cover letter writing; open access considerations; posting to AfricArXiv as a preprint; celebrating progress; reflecting on growth; next steps: conferences (IBRO, FENS, SfN, Pan African Neuroscience Congress), fellowships, further training; building your scientific identity as an African neuroscientist

Target journal submission portal; AfricArXiv for preprint

Submit final paper + 1-page reflection on the fellowship journey

SUBMIT DATA PAPER!

Appendix: Open African Datasets & Resources (incomplete)

Recommended African Electrophysiology Datasets

- OpenNeuro.org – search ‘Africa’ or ‘EEG’; filter by BIDS-compliant datasets
- EBRAINS (ebrains.eu) – African Brain Data Science Academy datasets
- African NeuroData Research Lab (ANR Lab) – datasets shared by ABDN partners
- Temple University EEG Corpus – large open EEG dataset useful for technique practice
- International Brain Laboratory (IBL) – large-scale electrophysiology datasets
- DANDI (dandiarchive.org) – Distributed Archives for Neurophysiology Data Integration

Key Free Learning Resources

- Neuroscience Online (neuroscience.uth.edu) – free open-access neuroscience textbook
- MNE-Python documentation & tutorials (mne.tools) – primary analysis tool
- OpenStax Neuroscience (free) – accessible foundational text
- BIDS Starter Kit (bids-standard.github.io) – FAIR data formatting guide
- AfricArXiv (africarxiv.org) – open preprint server for African research
- Neuropixels Training Course 2025 (UCL):
<https://www.ucl.ac.uk/brain-sciences/research/neuropixels/training/2025-neuropixels-course>
- Connected Papers (connectedpapers.com) – visual literature mapping tool
- Pynapple documentation (pynapple-io.github.io) – neural time series analysis

TA Notes

For Nicole & Happiness Eric

All session materials, datasets, and participant submissions should be stored in the shared ABDN Google Drive folder. Weekly sessions should begin with a 5-minute recap of the previous week. Encourage participants to use Google Colab if they experience local installation difficulties, this is common in African contexts with bandwidth or hardware constraints. For ethics sessions, be sensitive to varying institutional IRB processes across African countries. Guest speaker invitations should be confirmed at least 3 weeks in advance.